Full-Stack Web Development Architecture & React 19 Reference

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1. MODERN REACT 19 CONCEPTS

• **React Server Components (RSC):** Components executed solely on the server. Zero client bundle footprint for dependencies. Direct database and filesystem queries without intermediary API routes.
• **Server Actions & useActionState:** Async functions invoked from client forms with automatic optimistic updates via useOptimistic() and pending state transitions via useTransition().

2. REST vs GRAPHQL vs gRPC API ARCHITECTURE

• **REST:** Standard HTTP verbs (GET, POST, PUT, DELETE), resource-oriented URLs, standard status codes (200, 201, 400, 401, 404, 500), easy caching at CDN/browser level.
• **GraphQL:** Single endpoint (/graphql). Client defines exact query schema. Eliminates over-fetching and under-fetching. Ideal for complex graphs and mobile clients.
• **gRPC & Protocol Buffers:** Binary serialized RPC over HTTP/2 with bidirectional streaming. 7x-10x faster serialization than JSON. Standard for backend microservice communication.